FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

MIDTERM

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Name of Student	Name of Professor
Data Performed	Date Submitted

List the steps that you perform to come up with the program.

Paste the source code and explain the functionalities and take a screenshot of the final output.

```
section .data
filename db 'Receipt.txt', 0

Price db" A. Rice:Plain Rice       php 5.00",10
     db" B. Rice:Java Rice        php 10.00",10
     db" C. Ulam1: (Fish)         php 35.00",10
     db" D. Ulam2: (Pork)         php 50.00",10
     db" E. Ulam3: (Vegetables)   php 20.00",10
     db" F. Saging con yelo       php 60.00",10
PriceLen equ $ - Price

 Rice_Plain dd 5
 Rice_Java dd 10
 Ulam_Fish dd 35
 Ulam_Pork dd 50
 Ulam_Vegetables dd 20
 yelo dd 60

 TotalLabel db "Total Amount: Php "
TotalLabelLen equ $ - TotalLabel

NewLine db 10


ReceiptA db "Plain Rice - Php 5.00", 10
ReceiptALen equ $ - ReceiptA

ReceiptB db "Java Rice - Php 10.00", 10
ReceiptBLen equ $ - ReceiptB

ReceiptC db "Fish - Php 35.00", 10
ReceiptCLen equ $ - ReceiptC

ReceiptD db "Pork - Php 50.00", 10
ReceiptDLen equ $ - ReceiptD

ReceiptE db "Vegetables - Php 20.00", 10
ReceiptELen equ $ - ReceiptE
```

```
ReceiptF db "Saging con yelo - Php 60.00", 10
ReceiptFLen equ $ - ReceiptF




Menu db".....Georgette Eatery.....",10
    db": A. Rice:Plain Rice          :",10
    db": B. Rice:Java Rice           :",10
    db": C. Ulam1: (Fish)           :",10
    db": D. Ulam2: (Pork)            :",10
    db": E. Ulam3: (Vegetables)         :",10
    db": F. Saging con yelo          :",10

MenuLen equ $ - Menu


Confirmation db"Do you want to order again? [Y/N]: "

Confirmationlen equ $ - Confirmation

Thanks db"Thank you! Please come again!"

Thankslen equ $ - Thanks

%macro print_text 2
   mov ecx, %1
   mov edx, %2
   call print_proc
%endmacro

%macro print_choice 1
   mov eax, 3
   mov ebx, 0
   mov ecx, %1
   mov edx, 2
   int 0x80
%endmacro
```

```
%macro confirm_choice 1
mov eax, 3
   mov ebx, 0
   mov ecx, %1
   mov edx, 2
   int 0x80
%endmacro



section .bss

fileDescriptor resd 1

selectedLine resd 1

selectedLineLen resd 1

menu_choice resb 2


foodCount resd 1


; Accumulated total.
   totalFee resd 1
selectedPrice resd 1
continueInput resb 2



numberBuffer resb 12

global _start

section .text



_start:
```

```
;intialize total fee
 mov dword [totalFee], 0


mov eax, 5
   mov ebx, filename
   mov ecx, 577
   mov edx, 420
   int 0x80

   mov [fileDescriptor], eax

;an order loop for after confirming the yes option it goes back here
order_loop:

print_text Price, PriceLen

print_text Menu, MenuLen

read_menu_choice:
 print_choice menu_choice


 ; Plain Rice
   cmp byte [menu_choice], 'A'
   je choice_A
   cmp byte [menu_choice], 'a'
   je choice_A


   ; Java Rice
   cmp byte [menu_choice], 'B'
   je choice_B
   cmp byte [menu_choice], 'b'
   je choice_B

   ; Fish
   cmp byte [menu_choice], 'C'
   je choice_C
   cmp byte [menu_choice], 'c'
   je choice_C

   ; Pork
```

```
    cmp byte [menu_choice], 'D'
    je choice_D
    cmp byte [menu_choice], 'd'
    je choice_D

    ; Vegetables
    cmp byte [menu_choice], 'E'
    je choice_E
    cmp byte [menu_choice], 'e'
    je choice_E

    ; Saging con yelo
    cmp byte [menu_choice], 'F'
    je choice_F
    cmp byte [menu_choice], 'f'
    je choice_F

    jmp read_menu_choice

; they take the item name and price

;RecieptA
choice_A:
    mov eax, [Rice_Plain]
    mov [selectedPrice], eax
    mov dword [selectedLine], ReceiptA
    mov dword [selectedLineLen], ReceiptALen

    jmp accumulate_total

choice_B:
    mov eax, [Rice_Java]
    mov [selectedPrice], eax
     mov dword [selectedLine], ReceiptB
    mov dword [selectedLineLen], ReceiptBLen

    jmp accumulate_total

choice_C:
    mov eax, [Ulam_Fish]
    mov [selectedPrice], eax
    mov dword [selectedLine], ReceiptC
    mov dword [selectedLineLen], ReceiptCLen
```

```
    jmp accumulate_total

choice_D:
    mov eax, [Ulam_Pork]
    mov [selectedPrice], eax

    mov dword [selectedLine], ReceiptD
    mov dword [selectedLineLen], ReceiptDLen

    jmp accumulate_total

choice_E:
    mov eax, [Ulam_Vegetables]
    mov [selectedPrice], eax

     mov dword [selectedLine], ReceiptE
    mov dword [selectedLineLen], ReceiptELen

    jmp accumulate_total

choice_F:
    mov eax, [yelo]
    mov [selectedPrice], eax

     mov dword [selectedLine], ReceiptF
    mov dword [selectedLineLen], ReceiptFLen
    jmp accumulate_total



accumulate_total:
    ; Add selected price to total
    mov eax, [totalFee]
    add eax, [selectedPrice]
    mov [totalFee], eax

    ; Write selected food to receipt
    mov eax, 4
    mov ebx, [fileDescriptor]
    mov ecx, [selectedLine]
    mov edx, [selectedLineLen]
    int 0x80
```

```
    jmp confirmation_loop


confirmation_loop:

    print_text Confirmation, Confirmationlen

    confirm_choice continueInput

    ; Check uppercase Y
    cmp byte [continueInput], 'Y'
    je order_loop

    ; Check lowercase y
    cmp byte [continueInput], 'y'
    je order_loop

    ; Check uppercase N
    cmp byte [continueInput], 'N'
    je exit_program

    ; Check lowercase n
    cmp byte [continueInput], 'n'
    je exit_program

    ; Invalid input
    jmp confirmation_loop


exit_program:

    print_text Thanks, Thankslen

print_text NewLine, 1

    ; Write total label to receipt
    mov eax, 4
    mov ebx, [fileDescriptor]
    mov ecx, TotalLabel
    mov edx, TotalLabelLen
    int 0x80
```

```
    ; Convert totalFee into ASCII digits
    mov eax, [totalFee]
    mov ebx, 10
    mov edi, numberBuffer
    add edi, 11

    mov byte [edi], 10
    mov dword [foodCount], 1

convert_total:
    xor edx, edx
    div ebx

    add dl, '0'
    dec edi
    mov [edi], dl

    inc dword [foodCount]

    cmp eax, 0
    jne convert_total

    ; Write converted total to receipt
    mov eax, 4
    mov ebx, [fileDescriptor]
    mov ecx, edi
    mov edx, [foodCount]
    int 0x80

    ; Close receipt file
    mov eax, 6
    mov ebx, [fileDescriptor]
    int 0x80



print_proc:
    mov eax, 4
    mov ebx, 1
    int 0x80
    ret
```


@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $ nasm -f elf32 GeorgetteEatery.asm -o GeorgetteEatery.o
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $ ld -m elf_i386 GeorgetteEatery.o -o GeorgetteEatery
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $ ./GeorgetteEatery
A. Rice:Plain Rice php 5.00
B. Rice:Java Rice php 10.00
C. Ulam1: (Fish) php 35.00
D. Ulam2: (Pork) php 50.00
E. Ulam3: (Vegetables) php 20.00
F. Saging con yelo php 60.00
.....Georgette Eatery.....
: A. Rice:Plain Rice :
: B. Rice:Java Rice :
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :
A
Do you want to order again? [Y/N]: Y
A. Rice:Plain Rice php 5.00
B. Rice:Java Rice php 10.00
C. Ulam1: (Fish) php 35.00
D. Ulam2: (Pork) php 50.00
E. Ulam3: (Vegetables) php 20.00
F. Saging con yelo php 60.00

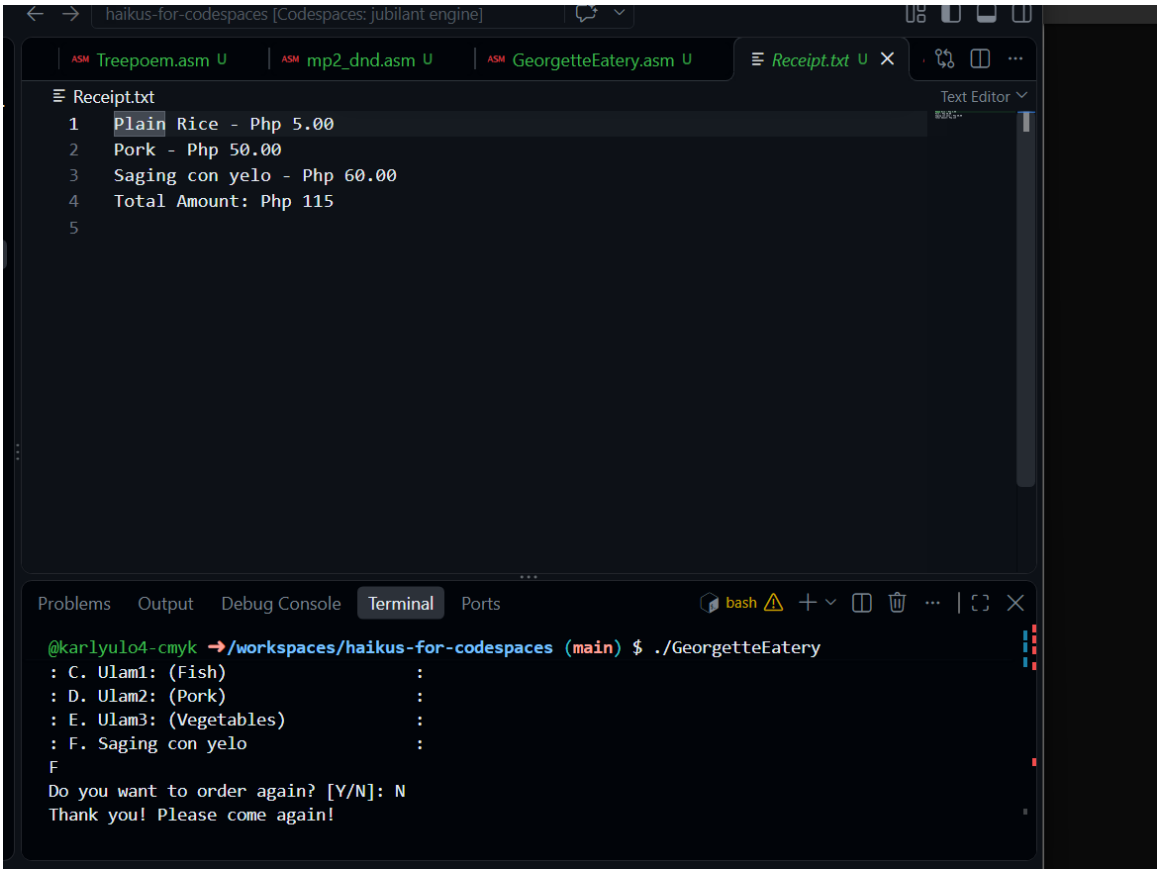
```
 F. Saging con yelo        php 60.00
...........Georgette Eatery...........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
A
Do you want to order again? [Y/N]: Y
 A. Rice:Plain Rice        php 5.00
 B. Rice:Java Rice         php 10.00
 C. Ulam1: (Fish)          php 35.00
 D. Ulam2: (Pork)          php 50.00
 E. Ulam3: (Vegetables)    php 20.00
 F. Saging con yelo        php 60.00
...........Georgette Eatery...........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
B
Do you want to order again? [Y/N]: Y
 A. Rice:Plain Rice        php 5.00
 B. Rice:Java Rice         php 10.00
 C. Ulam1: (Fish)          php 35.00
 D. Ulam2: (Pork)          php 50.00
 E. Ulam3: (Vegetables)    php 20.00
 F. Saging con yelo        php 60.00
...........Georgette Eatery...........
```

```
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
C
Do you want to order again? [Y/N]: Y
 A. Rice:Plain Rice        php 5.00
 B. Rice:Java Rice         php 10.00
 C. Ulam1: (Fish)          php 35.00
 D. Ulam2: (Pork)          php 50.00
 E. Ulam3: (Vegetables)    php 20.00
 F. Saging con yelo        php 60.00
..........Georgette Eatery..........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
D
Do you want to order again? [Y/N]: Y
 A. Rice:Plain Rice        php 5.00
 B. Rice:Java Rice         php 10.00
 C. Ulam1: (Fish)          php 35.00
 D. Ulam2: (Pork)          php 50.00
 E. Ulam3: (Vegetables)    php 20.00
 F. Saging con yelo        php 60.00
..........Georgette Eatery..........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
E
Do you want to order again? [Y/N]: Y
```


: F. Saging con yelo :
E
Do you want to order again? [Y/N]: Y
A. Rice:Plain Rice php 5.00
B. Rice:Java Rice php 10.00
C. Ulam1: (Fish) php 35.00
D. Ulam2: (Pork) php 50.00
E. Ulam3: (Vegetables) php 20.00
F. Saging con yelo php 60.00
.....Georgette Eatery.....
: A. Rice:Plain Rice :
: B. Rice:Java Rice :
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :
f
Do you want to order again? [Y/N]: N
Thank you! Please come again!
180
Segmentation fault (core dumped)
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $
[Preview] README.md
macro2.asm U
Treepoem.asm U
mp2_dnd.asm U
GeorgetteEatery.asm U
Receipt.txt U
FLAB
Receipt.txt
Text Editor
1 Plain Rice - Php 5.00
2 Java Rice - Php 10.00
3 Fish - Php 35.00
4 Pork - Php 50.00
5 Vegetables - Php 20.00
6 Saging con yelo - Php 60.00
7 Total Amount: Php 180
8
thanks
No results
Problems
Output
Debug Console
Terminal
Ports
bash
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $ ./GeorgetteEatery
: A. Rice:Plain Rice :
: B. Rice:Java Rice :
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :
f
Do you want to order again? [Y/N]: N
Thank you! Please come again!
180
Segmentation fault (core dumped)
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $


: A. Rice:Plain Rice :
: B. Rice:Java Rice :
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :
A
Do you want to order again? [Y/N]: Y
A. Rice:Plain Rice php 5.00
B. Rice:Java Rice php 10.00
C. Ulam1: (Fish) php 35.00
D. Ulam2: (Pork) php 50.00
E. Ulam3: (Vegetables) php 20.00
F. Saging con yelo php 60.00
.....Georgette Eatery.....
: A. Rice:Plain Rice :
: B. Rice:Java Rice :
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :

```
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (m
 F. Saging con yelo        php 60.00
...........Georgette Eatery...........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
D
Do you want to order again? [Y/N]: Y
 A. Rice:Plain Rice        php 5.00
 B. Rice:Java Rice         php 10.00
 C. Ulam1: (Fish)          php 35.00
 D. Ulam2: (Pork)          php 50.00
 E. Ulam3: (Vegetables)    php 20.00
 F. Saging con yelo        php 60.00
...........Georgette Eatery...........
: A. Rice:Plain Rice                  :
: B. Rice:Java Rice                   :
: C. Ulam1: (Fish)                    :
: D. Ulam2: (Pork)                    :
: E. Ulam3: (Vegetables)              :
: F. Saging con yelo                  :
F
Do you want to order again? [Y/N]: N
Thank you! Please come again!
```


haikus-for-codespaces [Codespaces: jubilant engine]
Treepoem.asm U
mp2_dnd.asm U
GeorgetteEatery.asm U
Receipt.txt U
Receipt.txt
Text Editor
1 Plain Rice - Php 5.00
2 Pork - Php 50.00
3 Saging con yelo - Php 60.00
4 Total Amount: Php 115
5
Problems
Output
Debug Console
Terminal
Ports
bash
@karlyulo4-cmyk ➜/workspaces/haikus-for-codespaces (main) $ ./GeorgetteEatery
: C. Ulam1: (Fish) :
: D. Ulam2: (Pork) :
: E. Ulam3: (Vegetables) :
: F. Saging con yelo :
F
Do you want to order again? [Y/N]: N
Thank you! Please come again!